

Arjuna JEE 2025

Maths

Sequence & Series

DPP: 5

Q1 If $|x| < 1$ then, the sum of the series $1 + 2x + 3x^2 + 4x^3 + \dots + \infty$ is

- (A) $\frac{1}{1-x}$
 (B) $\frac{1}{1+x}$
 (C) $\frac{1}{(1+x)^2}$
 (D) $\frac{1}{(1-x)^2}$

Q2 $\sum_{i=1}^{\infty} \frac{i}{7^i} =$

- (A) $\frac{7}{6}$ (B) $\frac{7}{36}$
 (C) $\frac{6}{7}$ (D) $\frac{36}{7}$

Q3 If $\frac{a^{n+1} + b^{n+1}}{a^n + b^n}$ is the AM of a and b , then n is equal to

- (A) 1 (B) -1
 (C) 0 (D) None of these

Q4 If $|3x - 1|, 3, |x - 3|$ are the first three terms of an arithmetic progression, then the sum of the first five terms can be

- (A) 5 (B) 10
 (C) 20 (D) 30

Q5 The four Arithmetic mean between 3 and 23 are

- (A) 5, 9, 11, 13
 (B) 7, 11, 15, 19
 (C) 5, 11, 15, 22
 (D) 7, 15, 19, 21

Q6 If the AM of the roots of quadratic equation is $\frac{8}{5}$ and AM of their reciprocal is $\frac{8}{7}$ then the equation is

- (A) $5x^2 - 16x + 7 = 0$
 (B) $7x^2 - 16x + 5 = 0$

(C) $7x^2 - 16x + 8 = 0$

(D) $3x^2 - 12x + 7 = 0$

Q7 If A_1, A_2 be two AM's and G_1, G_2 , be the GM between a and b then the value of $\frac{A_1 + A_2}{G_1 G_2} =$

- (A) $\frac{ab}{a+b}$
 (B) $\frac{a+b}{ab}$
 (C) $ab(a+b)$
 (D) $a^2 + b^2$

Q8 Let x be the arithmetic mean and y, z be two G.M between any two positive numbers then value of $\frac{y^3 + z^3}{xyz}$ is

- (A) 2
 (B) 3
 (C) $\frac{1}{2}$
 (D) $\frac{3}{2}$

Q9 If G_1 and G_2 are two geometric mean and A is the arithmetic mean inserted between two positive numbers a and b then the value of $\frac{G_1^2}{G_2} + \frac{G_2^2}{G_1}$ is

- (A) A
 (B) 2A
 (C) $\frac{A}{2}$
 (D) $\frac{3A}{2}$

Q10 If g_1, g_2 are two geometric mean between b and c then $g_1 g_2 =$

- (A) ac (B) ab
 (C) bc (D) abc

Q11



If a is the arithmetic mean between b and c and g_1, g_2 are two G. M between b and c then $g_1^3 + g_2^3 =$

- (A) abc
(B) $2abc$
(C) ab^2c
(D) abc^2

Q12 If ratio of AM and GM of two positive numbers a and b is $13 : 12$. Then the ratio $a : b$ is

- (A) $\frac{4}{9}$
(B) $\frac{9}{4}$
(C) $\frac{3}{2}$
(D) $\frac{2}{3}$

Q13 If A_1, A_2 be AM and G_1, G_2 be GM and H_1, H_2 are H. M between two positive quantity then $\frac{G_1 G_2}{H_1 H_2} =$

- (A) $\frac{A_1 + A_2}{H_1 + H_2}$
(B) $\frac{H_1 + H_2}{A_1 + A_2}$
(C) $\frac{G_1 + G_2}{H_1 + H_2}$
(D) $\frac{A_1 + A_2}{G_1 + G_2}$

Q14 If H is the harmonic mean between P and Q

then the value of $\left(\frac{H}{P} + \frac{H}{Q}\right)$ is

- (A) 2
(B) $\frac{PQ}{P+Q}$
(C) $\frac{P+Q}{PQ}$
(D) None

Q15 After inserting n , AMs between 2 and 38 the sum of the resulting progression is 200. The value of n is

- (A) 10
(B) 8
(C) 9
(D) None of the above

Q16 The harmonic means of the roots of the equation $(5 + \sqrt{2})x^2 - (4 + \sqrt{5})x + 8 + 2\sqrt{5} = 0$ is

- (A) 2 (B) 4
(C) 6 (D) 8

Q17 If the harmonic mean between roots of $(5 + \sqrt{2})x^2 - bx + 8 + 2\sqrt{5} = 0$ is 4 then b

- (A) 2
(B) $4 - \sqrt{5}$
(C) 3
(D) $4 + \sqrt{5}$

Q18 Between two numbers whose sum is $\frac{13}{6}$ an even number of AM are inserted. If the sum of the means exceeds their numbers by unity. Then number of means inserted equal to

- (A) 6
(B) 12
(C) 18
(D) None of the above

Q19 Between 1 and 31, n AM are inserted such that the ratio of the 7^{th} and $(n - 1)^{\text{th}}$ mean is $5 : 9$ then $n =$

- (A) 12 (B) 13
(C) 14 (D) 15

Q20 After inserting n A.M's between 2 and 38, the sum of the resulting progression is 200. The value of n is

- (A) 10 (B) 8
(C) 9 (D) none of these

Q21 If $2, h_1, h_2, \dots, h_{20}, 6$ are in harmonic progression and $2, a_1, a_2, \dots, a_{20}, 6$ are in arithmetic progression, then the value of $a_3 h_{18}$ is equal to

- (A) 6 (B) 12
(C) 3 (D) 9



Answer Key

Q1 (D)

Q2 (B)

Q3 (C)

Q4 (A)

Q5 (B)

Q6 (A)

Q7 (B)

Q8 (A)

Q9 (B)

Q10 (C)

Q11 (B)

Q12 (B)

Q13 (A)

Q14 (A)

Q15 (B)

Q16 (B)

Q17 (D)

Q18 (B)

Q19 (C)

Q20 (B)

Q21 (B)

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